Market-Level Capability Gap Analysis

Executive Summary

The global knowledge workforce is operating generative AI tools at scale without the foundational understanding required to use them accurately or evaluate their outputs reliably. Eighty-six percent of employers expect AI to transform their businesses by 2030, 63% identify workforce skill gaps as their primary barrier to business transformation, and half of executives name missing skills as the top barrier to AI adoption (WEF 2025, pp. 49, 62). The gap between individual and organizational return is now measured directly. Eighty percent of respondents to McKinsey's 2026 global survey report that AI improved their own productivity. The share reporting any earnings impact at their organization stays at 37% (Tinkoff et al., 2026, pp. 3–4). At the individual level, professionals report broad time savings from AI use while simultaneously demonstrating reduced critical engagement with AI-generated outputs, a divergence with direct consequences for output quality, compliance exposure, and organizational productivity. The adoption gap is not evenly distributed: geographic disparities in AI use mean that organizations with internationally distributed teams face internal competency imbalances on top of market-level skill shortfalls. A further structural barrier compounds the skill gap: 69% of professionals who use AI tools conceal that use from colleagues, a dynamic that individual training programs alone cannot resolve. Existing solutions (vendor-branded tool certifications, platform content libraries, and custom enterprise programs) address tool operation rather than evaluative judgment, measure completion rather than behavior change, and ignore the social conditions that determine whether individual skill translates into organizational practice. This analysis documents the market context, performance gap, business impact, change management dimension, and limitations of current solutions that define the design space for a structured, competency-based AI literacy program targeting the broad mid-career knowledge workforce.

1. Market Context

The generative AI adoption curve has moved from early-adopter experimentation to broad workforce expectation faster than organizational capability has kept pace. Enterprise change also lands slowly. Evans's standing comparison is public cloud. After more than a decade it held about 30% of enterprise workloads, according to the Goldman Sachs CIO Survey he reproduces (Evans, November 2025, slide 58). Since the release of ChatGPT in November 2022, investment flows into AI have increased nearly eightfold (WEF 2025, p. 11). The supply side has kept spending. Company guidance compiled by Evans puts the four largest hyperscalers' planned 2026 capital expenditure at about $700 billion (Evans, May 2026, slide 8). Global telecoms, by comparison, spend about $300 billion. The gap between that spending and the deployment figures below is the market-level version of the capability

gap this program addresses. Investment intent has not slowed. Sixty percent of respondents expect their organization to raise AI spending in the coming year. The survey authors note that conviction in AI grows faster than the returns organizations can attribute to it (Tinkoff et al., 2026, pp. 16, 29). Diffusion history supports the lag. As late as 2023, only about one in five companies ran most applications in the cloud, a technology available since the mid-2000s (Yee et al., 2025, p. 7). The organizational response is already visible in employer planning data: 86% of employers expect AI and information processing technologies to transform their business by 2030, with Financial Services (97%) and Electronics (95%) sectors above the global average (WEF Future of Jobs Report 2025, p. 62). Upskilling the workforce is the most commonly anticipated response, with 85% of surveyed employers planning to adopt this strategy over the 2025–2030 period (WEF 2025, p. 52). Demand for AI-specific capability is already current, not projected. AI and big data top the list of the fastest-growing skills globally, followed by networks and cybersecurity and technological literacy (WEF 2025, p. 37, Figure 3.4).

Adoption, however, is neither uniform nor matched by competency, and the disparities that exist across geographic boundaries have direct consequences for multinational organizations. Anthropic's Economic Index documents that Singapore uses Claude at 4.6 times the rate expected given its working-age population; Canada at 2.9 times. Indonesia, by contrast, sits at 0.36 times expected usage; India at 0.27 times; Nigeria at 0.2 times (Appel, McCrory, and Tamkin 2025, p. 3). A second dataset points the same way from the developer side. On OpenRouter, a routing platform serving 300-plus models, Asia's share of spend more than doubled from about 13% to about 31% between late 2024 and late 2025. Singapore alone accounted for 9.2% of token volume, second only to the United States (OpenRouter and a16z, 2025, pp. 34–36). The authors caution that region is assigned by billing location, so Singapore's share may reflect billing entities rather than users. Employer intent in the high-usage economies is already at ceiling. Ninety-seven percent of employers in Singapore plan to prioritize upskilling and 82% plan to reskill staff to work alongside AI, yet they expect 70 of every 100 workers to need training by 2030, against 59 globally; 58% still expect to prioritize university degrees in hiring (WEF 2025, pp. 67, 192–193). In Canada, 97% expect AI to transform their operations, 90% plan reskilling alongside AI, and employers expect 68 of 100 workers to need training (WEF 2025, pp. 74, 124–125). For an organization operating teams across Singapore, Jakarta, and Mumbai, this is an internal capability gap, not merely a market-level observation. Teams in the same organization, collaborating on the same deliverables, may be operating AI tools at markedly different rates and levels of fluency. Without a standardized competency baseline, distributed workforces inherit the global adoption disparity as an internal productivity and equity problem. A structured AI literacy program that establishes shared capability across geographic contexts puts internationally distributed teams on common professional footing.

Within enterprise deployment more broadly, institutional inertia and fixed adoption costs concentrate early AI use among specialized tasks where deployment is straightforward, capabilities are robust, and economic returns are clear, a pattern that systematically excludes the broad

knowledge workforce from the productivity gains that AI adoption makes possible (Appel, McCrory, and Tamkin 2025, pp. 30–31). Morgan Stanley's CIO survey, reproduced by analyst Benedict Evans (November 2025, slide 57), frames this structurally. By September 2025 about a quarter of CIOs had deployed at least one LLM project in production. About 40% planned none before 2026 or later. The March 2025 wave of the same survey (Evans, May 2025, slide 44) showed the same split, so six months of deployment moved neither figure. By mid-2026 the picture had moved but not resolved. Among organizations using AI, 22% were still experimenting, 34% piloting, and 44% scaling across the enterprise, up from 38% a year earlier (Tinkoff et al., 2026, p. 5). Enterprise usage records through March 2026 show who moved first. Among U.S. public companies, ChatGPT Enterprise adopters had a median workforce of 2,934 against 424 for non-adopters (Chatterji et al., 2026, p. 11). Firms in the top revenue quartile were 6.9 percentage points more likely to adopt (p. 14). Adoption tracked prior stocks of R&D, SG&A, and capitalized software rather than scale alone (p. 15). The authors warn that this pattern could widen productivity differences across firms even when the models are broadly available (p. 22). Their estimates cover large public firms and say nothing about small private firms or mid-market companies (p. 14). The bottleneck is not hardware, licensing, or access. It is the organizational and human infrastructure required to convert tool availability into reliable, judgment-informed use at scale. Two McKinsey analyses reach the same point from opposite directions. The 2026 enterprise survey finds the limiting factor is the organization's ability to absorb change (Tinkoff et al., 2026, p. 24). MGI's task-level model finds that applying AI to individual tasks inside legacy workflows is unlikely to deliver the available productivity gains (Yee et al., 2025, p. 35). Employers across 52 of 55 surveyed economies and 19 of 22 sectors rank workforce skill gaps as the leading barrier to business transformation, up from 60% to 63% since the 2023 edition of the same report (WEF 2025, p. 49).

2. Documented Performance Gaps

The gap between AI tool access and productive AI use is measurable at the individual level. The Anthropic Interviewer study (December 2025), drawing on qualitative and quantitative data from 1,250 professionals (1,000 from the general workforce, 125 scientists, 125 creatives), is the largest interview sample on this question to date. Three limits apply. Claude conducted the interviews, so participants discussed AI with an AI. The post serves product development as well as research. And its authors qualify the headline finding as holding 'in our sample.' The findings are self-reports, not observed behavior.

At the aggregate level, adoption is broadly positive on self-reported outcomes: 86% of professionals report that AI saves them time, and 65% report satisfaction with the role AI plays in their work. These figures confirm that tools are being used and perceived as valuable. They do not confirm that tools are being used well.

The performance gap is located not in adoption rates but in judgment quality. The Anthropic Interviewer data reveals a meaningful divergence between how professionals characterize their AI use and what task-level behavioral data shows. Sixty-five percent of participants describe AI's primary role in their work as augmentative: the model supports their judgment rather than replaces it. Anthropic's task-level analysis of actual Claude usage, however, shows a more even split: 57% augmentative, 43% automative (Handa et al. 2025, p. 3). The size of that gap depends on the baseline: eight percentage points against Handa et al., widening to eighteen against the more recent Economic Index wave the Anthropic Interviewer cites (August 2025 data, approximately 47% augmentative; Appel, McCrory & Tamkin, 2025, p. 11), an 8-to-18-point overestimate of augmentative use depending on the dataset (Handa et al. 2025, p. 3; Anthropic Interviewer, Dec 2025). The discrepancy suggests that a meaningful share of users believe they are exercising oversight over AI outputs when the task structure indicates otherwise, a blind spot with direct consequences for output quality and compliance exposure.

The critical thinking research compounds this finding. Lee et al. (CHI 2025), surveying 319 knowledge workers across 936 real-world AI-assisted work tasks, found that higher confidence in generative AI is associated with less critical thinking, while higher confidence in one's own ability to do the task without AI is associated with more (Lee et al. 2025, p. 9). The authors measured task-specific self-confidence, not domain expertise as such, and only speculate that building expertise would help (p. 14). The study identifies a pattern of overreliance (users reduce scrutiny of AI outputs precisely when their confidence in the tool is highest) and finds that barriers to critical engagement include time pressure (44 of 319), limited motivation when tasks are perceived as trivial (55 of 319) or when the tool is trusted (83 of 319), and difficulty evaluating AI responses without enough domain knowledge (58 of 319) (Lee et al. 2025, p. 11). An institutional survey in 2026 found the same shape. Respondents reported that the temptation to offload thinking peaked under deadline pressure and that they felt increasingly dependent on AI (MIT Ad Hoc Committee, 2026, pp. 7, 32). The most common written-in concern was that reliance could erode critical thinking and long-term skill (p. 33). This is the structural risk that tool familiarity without foundational model understanding creates: a workforce that uses AI frequently, reports satisfaction, and reduces scrutiny of outputs in proportion to how comfortable they feel with the tool.

The productivity data from Tamkin and McCrory (2025) establishes what is at stake when this gap is closed. Across 100,000 real-world Claude conversations, AI reduces median task completion time by 81%, with the distribution concentrated in the 50–95% range and peaking between 80–90% (Tamkin and McCrory 2025, pp. 12–13). Specific task categories show sharper variation: compiling information from reports yields approximately 95% time savings; document drafting, 87%; financial analysis tasks, 80% (Tamkin and McCrory 2025, pp. 10, 12). These figures derive from observational analysis of Claude conversations and sit at the optimistic end of the evidence; the randomized controlled trials referenced in the same work report more conservative task-time reductions in the 14–56% range. The program treats the 81% median as a capability-contingent ceiling rather than a baseline expectation:

the distance between the observational median and the controlled-trial range is itself a function of user skill. Translating task-level efficiencies to macroeconomic terms, the analysis implies a 1.8% annual increase in US labor productivity over the next decade under broad adoption, double the growth rate observed since 2019 (Tamkin and McCrory 2025, p. 3). Anthropic's January 2026 report replicates the 1.8% on a larger sample and then discounts each task's speedup by its measured success rate, which lowers the implied gain to 1.2 points for consumer use and 1.0 for enterprise API use (Appel, Massenkoff, and McCrory 2026, p. 48). That potential is contingent on workforce capability, not tool deployment alone. Context constrains sophisticated use: curating appropriate context for AI models is critical for high-impact deployments, and the bottleneck for many organizations is not model capability but the data and organizational infrastructure needed to give the model the context complex work requires (Appel, McCrory, and Tamkin 2025, pp. 5, 41, 44); the human side of that infrastructure is the capability this program addresses.

The gap, stated precisely: tools are deployed; productivity gains are documented at the task level; but the workforce lacks the evaluative judgment to verify outputs, the conceptual foundation to understand model failure modes, and the competency vocabulary to apply these skills consistently across tasks and contexts.

3. Business Impact

The business consequences of the documented performance gap operate across three dimensions: direct productivity loss from underutilization, quality and compliance risk from uncritical output acceptance, and structural competitive disadvantage as AI-capable peers accelerate.

Productivity underutilization. Tamkin and McCrory (2025), using O*NET occupational classifications matched to BLS wage data, estimate that the median AI-assisted task would cost $54 in professional labor to complete unassisted (the worker's hourly wage multiplied by the task's unassisted duration). By occupation category, business and financial tasks average $69; management tasks, $133; computer and mathematical tasks, $82 (Tamkin and McCrory 2025, p. 10). Time savings convert directly into recovered labor cost. At the 81% observational median, a $133 management task takes roughly $25 of labor to complete with proficient AI use, recovering about $108 in labor value each time the task occurs; at the more conservative 14–56% controlled-trial range, the same task still recovers between roughly $19 and $74. The underutilization cost is the share of that recoverable value a worker forfeits through surface-level engagement: single-turn prompting, uncritical output acceptance, or avoidance of AI for tasks where judgment concerns create hesitation. The forfeited value recurs with every occurrence of every AI-amenable task, across every knowledge worker in the organization. The 1.8% labor productivity implication from Tamkin and McCrory (2025) is a macroeconomic estimate under broad, proficient adoption; the organizational delta between proficient and non-proficient use is the addressable gap a competency program closes.

Quality and compliance risk. Lee et al. (CHI 2025) document that confidence in AI is associated with less critical engagement and that trust in the tool leads users to accept outputs by default, especially where they lack the domain knowledge to inspect them (pp. 11, 14). For knowledge workers in functions where output accuracy carries legal, financial, or reputational stakes (compliance, finance, legal, HR, product), this reduction in scrutiny represents a material risk exposure. The WEF (2025, p. 11), summarizing the field-experiment literature it cites, notes that generative AI enhances knowledge work within its capability and risks adverse outcomes where users unknowingly stretch it beyond that capability; its own skills analysis adds that human oversight remains crucial even where the tool can assist (p. 44). Without a framework for recognizing those capability boundaries (for instance, that a model generating a plausible regulatory citation may be confabulating rather than retrieving), knowledge workers in high-stakes functions have no reliable basis for determining when AI output requires independent verification and when it does not. The concern is not confined to the workplace. MIT's 2026 review names hallucinated citations as a specific risk to train against and reports hallucination as respondents' leading concern about AI (MIT Ad Hoc Committee, 2026, pp. 22, 33). It makes the submitter responsible for any inaccurate content produced with AI assistance (p. 16).

Competitive positioning. Skill gaps in the labor market are the primary barrier to business transformation cited by employers for the 2025–2030 period, named by 63% of respondents, a figure that has increased since the 2023 report and ranks first across 52 of 55 economies and 19 of 22 sectors surveyed (WEF 2025, p. 49). Organizations that close the individual competency gap ahead of competitors gain compounding advantage: their workforce applies AI earlier, more accurately, and across a broader range of tasks. McKinsey's 2026 survey profiles the 6% of respondents whose organizations attribute at least 5% of earnings to AI. Compared with the rest, they are about three times as likely to have redesigned workflows (74% against 25%). They are twice as likely to have defined processes for measuring AI's impact and twice as likely to report visible senior-leader commitment. They are 3.3 times as likely to intend to transform their business within three years (Tinkoff et al., 2026, pp. 17–19). Individual competency is the input those practices act on. Without them, the survey finds, individual gains stay individual. WEF respondents project that by 2030, the share of tasks performed by humans alone will decline from 47% toward near-parity with machine and human-machine collaboration categories (WEF 2025, p. 26), a structural shift that rewards organizations whose workforces can navigate the human-machine interface with sound judgment, not merely operate the tools on one side of it.

The business case is a quantifiable comparison between the productivity trajectory of a competency-developed workforce and one that remains at surface-level adoption. The difference in trajectory is measurable in current data.

4. Change Management Context

Training design alone cannot close the documented performance gap. The Anthropic Interviewer (2025) identifies a behavioral dynamic that operates independently of, and in some cases contrary to, individual skill acquisition: 69% of professionals surveyed report that social stigma is an active barrier to AI use at work. The finding is not that workers reject AI tools (adoption rates and self-reported time savings contradict that reading) but that a majority of current AI users conceal their usage from colleagues. One participant's account is representative: a fact-checker reported that when a colleague expressed hostility toward AI, they said nothing and did not disclose their own practice, because they knew how others in their team felt.

The mechanism is concealment, not resistance. Employer planning data explains why concealment is a rational response rather than a character flaw. Forty-one percent of employers expect to downsize where AI can replicate people's work, and 47% plan to transition staff out of AI-disrupted roles (WEF 2025, p. 63). The same employers rank organizational culture and resistance to change as the second barrier to transformation (46%, p. 49), and 43% of executives name lack of vision among managers as a barrier to AI adoption (p. 62). The gap is organizational on both sides of the manager's desk. Workers who have individually adopted AI tools, who report satisfaction and time savings, and who would be candidates for peer knowledge-sharing instead operate in isolation, each navigating AI use privately, without shared vocabulary, organizational norms, or social permission to make their practice visible. This dynamic has two direct consequences for program design. First, a competency program that addresses only individual skill acquisition leaves the organizational conditions that suppress transfer entirely unaddressed. The perception gap is not only between what workers say and do; it also runs between what employers plan and what workers value. In the WEF's comparison with ADP Research's employee survey, employers rank well-being first and upskilling second among talent practices, employees rank them eighth and seventh, and employees put working hours first where employers put it eighth (WEF 2025, pp. 53–54). A program that assumes workers want the training their employer plans is building on the employer's half of that gap. A worker who completes a program and returns to a team culture where AI use signals laziness or replaceability is no more likely to visibly apply those skills than before. Second, the absence of shared professional vocabulary makes peer normalization impossible. Workers cannot calibrate their AI use against colleagues, identify best practices through informal observation, or develop team-level norms without language that makes AI work discussable in the first place.

This is why the 4D Framework (Delegation, Description, Discernment, Diligence; Dakan & Feller, 2025) functions not only as a competency taxonomy but as a change management instrument. Its authors present it as a tool to inform discourse and practice, compatible with other skills taxonomies rather than a replacement for them (Dakan & Feller, 2025, pp. 1, 3). A workforce that shares common vocabulary for AI collaboration can name what they are doing, discuss where they draw delegation boundaries, and surface discernment concerns without the binary framing of "pro-AI" or "anti-AI"

that makes current workplace conversations about AI tools socially fraught. The program addresses the change management dimension not through a separate communications strategy but through the design of the learning experience itself: shared language, normalized peer practice, and visible demonstration that evaluative judgment, not uncritical acceptance, is the professional standard.

Mollick (2012, *Strategic Management Journal*), analyzing 854 PC games from 395 companies with just over $4 billion in combined revenue, found that individual-level differences account for more variance in performance than organizational-level factors. Individual producers, the mid-level project managers, explained 22.3 percent of the variance in product revenue and designers 7.4 percent, against 21.3 percent for the firm itself. His applied work, *Co-Intelligence* (2024), extends this finding to AI adoption: the most advanced enterprise AI users typically operate without institutional authorization and self-direct their adoption because top-down rollouts optimize for compliance rather than capability. *Co-Intelligence* is a practitioner text rather than a peer-reviewed empirical study; its directional argument is consistent with both the 2012 findings and the Anthropic Interviewer's stigma data: the workers most capable of generating AI-enabled performance returns are already using AI privately, outside organizational visibility, precisely because the organizational apparatus has not created conditions for that use to be normalized. A program that closes the individual competency gap while establishing shared professional vocabulary addresses both the skill deficit and the concealment dynamic simultaneously.

5. Existing Solutions and Their Limitations

Market Structure

The enterprise AI training market in 2025–2026 organizes into three tiers: vendor-branded programs that subsidize product adoption, platform content libraries that aggregate self-paced courses, and enterprise custom solutions deployed through organizational L&D functions. Each tier has a distinct design philosophy, a distinct delivery model, and a consistent structural limitation.

Tier 1 — Vendor-Branded, Tool-Centric Programs

The dominant programs in this category are produced and distributed by the technology companies whose products they promote. Three offerings define the tier.

Microsoft Career Essentials in Generative AI (LinkedIn Learning, developed with Microsoft) consists of five modules covering generative AI concepts, Microsoft 365 Copilot operation, prompt engineering within the Microsoft ecosystem, and ethics. The program was offered free through 2025 and awards a Professional Certificate upon completion; its content is explicitly oriented toward Copilot rather than generative AI across tools or contexts. The course is high-volume (over 700,000

learners have enrolled in the ethics module alone), but its design logic is promotional. A learner who completes the program can operate Copilot more effectively; they are not equipped to evaluate AI-generated output for accuracy, understand why an LLM produces a plausible-sounding but factually incorrect claim, or transfer their skills when Copilot is replaced or supplemented by a different system.

Google AI Essentials Specialization (Coursera, ~10 hours) covers foundational AI concepts, prompting, and responsible AI with exercises integrated into Google Workspace. Independent reviewers describe the course as approachable and well-structured for non-technical learners, particularly the prompting module, but consistently identify it as surface-level: minimal depth on advanced techniques, real-world case studies, or implementation strategies. Like the Microsoft program, it teaches tool operation within a specific vendor ecosystem and provides no mechanism for ensuring that skills transfer to the learner's actual work context. Google subsequently released a Google AI Professional Certificate (Coursera, 2026) extending coverage across six functional domains, but the foundational course, the entry point for the majority of learners, retains the same surface-level design.

Anthropic's AI Fluency Suite is the most conceptually sophisticated offering in this tier and the broadest in scope. The suite comprises five courses: *AI Fluency: Framework and Foundations* (2025), organized around the 4D competency framework; *AI Capabilities and Limitations* (2026), covering the behavioral properties of generative AI models; *Teaching AI Fluency* (2025), designed for educators integrating AI literacy into formal instruction; *AI Fluency for Non-Profits* (2025), an audience-specific adaptation of the core framework; and *Claude 101* (2025), a practical tool-use guide that, like the Microsoft and Google programs, is explicitly product-centric. The suite spans a wider range of contexts and pedagogical purposes than any competitor offering in this tier, and several of its courses are explicitly designed to transfer across tools rather than train learners on a single platform.

Within this suite, the *AI Capabilities and Limitations* course represents a substantive advance. It frames generative AI around four behavioral properties (Next Token Prediction, Knowledge, Working Memory, and Steerability) and explicitly addresses hallucination mechanics: "a citation that *looks* like a real citation satisfies the pattern just as well as one pointing to a paper that actually exists" (Anthropic 2026). The course teaches learners to locate tasks on a capability-limitation continuum and to treat specificity (names, dates, statistics, citations) as the zone where fabrication concentrates and verification is most warranted. This is real foundational content that most competitor offerings do not approach.

Two limitations remain relevant to the present analysis. First, the course addresses next-token prediction as a *behavioral* property: what AI outputs look like, when to scrutinize them, how to interpret confident-sounding errors. It does not address tokenization as a *mechanical process*: how text is parsed into subword fragments before prediction begins, why those boundary decisions produce systematic errors in arithmetic, character-level tasks, and non-English text, and why those failure

modes are structurally predictable rather than random. The behavioral framing is valuable and actionable; the mechanical layer is the foundation that makes the behavioral framing explicable rather than rote. Second, the course's value is contingent on active engagement with its exercises, hands-on tasks designed to produce genuine calibration against real work. The platform does not enforce that engagement. A learner who clicks through without completing the exercises exits with terminology rather than capability, a distinction the course design cannot prevent and the assessment structure does not catch. The distinction has support beyond instructional-design convention. Lee et al. (CHI 2025, p. 3) summarize prior work showing that cognitive forcing functions, such as requiring the user to act on an output before proceeding, reduce overreliance more than explanations do, and their own recommendations call for interventions that build ability rather than add a per-task audit (pp. 14–15). This program's twelve practice activities are that forcing function; a video-and-quiz course is the explanation.

Both courses carry a 2026 copyright; the *AI Capabilities and Limitations* course published its first video content in April 2026. The skill deficiencies captured in the WEF Future of Jobs Report 2025, the Anthropic Economic Index reports, and the Anthropic Interviewer findings reflect a workforce that entered the generative AI era without access to these resources. The gap existed before the most sophisticated available solution did. Anthropic's own workforce study closes the loop: after documenting skill-atrophy concerns and a 'paradox of supervision' among its engineers, the company says its internal follow-up on best practices for AI-augmented work will be guided by the same fluency framework, and that it is supporting CodePath's adaptation of computer science curricula (Huang et al., 2025). The framework is therefore not only a public course scaffold but the vendor's stated internal remedy for problems it measured in its own staff.

Tier 2 — Platform Content Libraries

Coursera, LinkedIn Learning, and DataCamp host extensive catalogs of AI-related courses from IBM, DeepLearning.AI, CompTIA, and independent instructors. DeepLearning.AI's *AI for Everyone* (Andrew Ng) provides a non-technical overview of AI capabilities and use-case identification; IBM's *AI Foundations for Everyone* covers conceptual underpinnings with some no-code tool practice; Google AI Essentials prioritizes practical generative AI productivity tasks for non-technical learners.

CompTIA's *AI Essentials* (launched November 2025) represents a more applied design than earlier platform offerings, targeting knowledge workers specifically and adding scenario-based prompting exercises, use-case identification, and data security and privacy fundamentals. CompTIA's own research, however, documents that just 34% of companies currently require AI skills training for their employees. Platform availability has not translated into systematic enterprise deployment.

Organizations with mature, workforce-wide AI upskilling programs are nearly twice as likely to report significant positive AI ROI as those without: the share reporting strong returns rises from approximately 22% to 42% when structured capability building accompanies AI tool investment

(DataCamp 2026 State of Data and AI Literacy Report; YouGov, n = 500+ US and UK enterprise leaders). That figure is from a vendor-commissioned survey and should be read accordingly; the directional finding is consistent with independent transfer research. What that same research confirms is that while 82% of enterprise leaders say their organization provides some form of AI training, only 35% report having a mature, workforce-wide upskilling program. Content library access and organizational capability are not the same thing.

The structural limitation of platform programs is that completion is self-directed and organizationally disconnected. A 95% completion rate confirms that employees clicked through modules; it does not confirm that they retained information, applied it on the job, or improved performance in any measurable respect.

Tier 3 — Enterprise Custom and Consulting Solutions

A smaller market segment offers custom-built programs deployed through internal L&D functions, often with vendor support. This tier produces stronger outcomes when implemented well, as the DataCamp data above suggests. The structural limitation is access: custom solutions require internal L&D capacity, budget allocation, and executive sponsorship that many organizations lack. Seventy-four percent of organizations report not keeping up with their company's demand for new skills despite $400 billion in annual training expenditure (Bersin, as cited in Hardman 2026), a figure consistent with Hardman's (2026) synthesis of the transfer research literature, which places sustained behavior change rates from formal training at 10–20%. High-quality bespoke design does not scale by default.

Structural Limitations Across the Market

Three design failures appear consistently across all three tiers.

1. Tool familiarity without foundational model understanding. The dominant design paradigm across Tiers 1 and 2 trains learners to operate specific AI interfaces without establishing the conceptual model required to evaluate outputs, recognize failure modes, or transfer skills across tools and contexts. This is also the gap Lee et al. (CHI 2025, p. 15) identify: generative AI creates new cognitive tasks, above all integrating and stewarding output, and training that teaches tool operation does not address them. The WEF (2025, p. 11) notes that generative AI risks producing adverse outcomes specifically where users unknowingly stretch the technology beyond its capability, a risk that tool-use training does not equip learners to recognize or manage. A learner who can generate a Copilot summary or a Gemini draft cannot necessarily determine whether the output is accurate, where it may have introduced hallucination, or when a task should not be delegated to an AI system at all. Anthropic's *AI Capabilities and Limitations* course comes closest to addressing this, and represents a genuine step toward transferable foundational understanding. It does not, however, address the mechanical layer (tokenization, subword parsing, the structural predictability of certain

failure modes) that would allow learners to reason from first principles about where AI judgment applies and where it does not. The behavioral framing teaches learners what to do; the mechanical foundation explains why.

2. No behavior transfer mechanism. The most reliable indicator of effective AI training is task-level behavioral change, not course completion. Teams that apply AI tools to live tasks within 48 hours of training retain skills at significantly higher rates; role-specific training produces meaningfully higher adoption than general AI literacy programs. None of the reviewed programs include a 30/60/90-day behavioral follow-up, manager observation protocol, or structured reinforcement loop. The pattern predates the corporate market. A 2023 systematic review of AI-literacy interventions found hardly any that assessed whether participants understood AI concepts afterward (Casal-Otero et al., 2023, as reviewed in Ritchot, 2025, p. 7). The consequence follows directly from Hardman's (2026) synthesis: when only 10–20% of formal training leads to sustained behavior change, programs that are passive, decontextualized, and completion-tracked will cluster at the low end of that range by design.

3. No change management dimension. The 69% social stigma finding from the Anthropic Interviewer (2025) represents an adoption barrier that no course completion strategy resolves. CompTIA has publicly acknowledged the need to consider the change management aspects of AI adoption alongside formal training investment, but program design across all three tiers addresses this concern rhetorically at best. None of the reviewed offerings include structural features (shared vocabulary, team-level norms, manager activation) that create the social conditions for normalized AI use.

The Individual Competency Design Imperative

The cumulative picture the market presents is not one of inadequate supply. It is one of systematic misalignment between what the research identifies as the performance gap and what available programs are designed to close. As the change management analysis established, individual-level differences drive firm performance, and the workers most capable of generating AI-enabled returns are already self-directing their adoption privately, outside organizational visibility (Mollick 2012; Anthropic Interviewer, Dec 2025).

The structurally correct intervention targets the individual knowledge worker (their judgment, their evaluative vocabulary, their transfer of skill across tools and contexts) rather than organizational-level compliance protocols or executive-level strategy. Programs designed at the organizational level manage the risk of AI adoption. Programs designed at the individual competency level generate the performance return that makes adoption worthwhile. The two are not substitutes. McKinsey's 2026 survey finds individual gains already widespread (80%) and enterprise returns flat (37%). It attributes the difference to workflow redesign and leadership practice rather than to tool insertion (Tinkoff et al., 2026, pp. 4, 14, 18). Its authors frame growing employee AI fluency as the resource leaders should use to redesign end-to-end workflows (p. 14). This program supplies the fluency and the shared

vocabulary; the Level 3 and Level 4 instruments are where the organizational practices attach. The current market offers risk management at scale. Individual competency with an organizational transfer path remains a design gap. The competency set is converging across independent designers. MIT's 2026 committee, working without reference to Anthropic's framework, defined effective use in four parts: how to specify a prompt, how to verify an output, when a model is likely to hallucinate, and when not to reach for AI at all (MIT Ad Hoc Committee, 2026, p. 21). It defined responsible use as distinguishing augmentation from automation and disclosing AI contributions honestly (p. 21). Those are Description, Discernment, Delegation, and Diligence under other names.

Sources

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